



St Catherine's
School
Waverley, Sydney

Student Number:

Year 12
Assessment Task 1
2010

Mathematics Extension II

Time allowed: 55
minutes

Course weighting:
15%

General Instructions

- Attempt ALL questions
- Write your
Student NUMBER at the
top of this page and on
each writing booklet used

Sections

Marks

Total marks

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

Note: $\ln x = \log_e x, \quad x > 0$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Question 1.

a) Let $z=1+2i$ and $w=1+i$ find in the form $x+iy$

(i) $z\bar{w}$ 1m

(ii) $\frac{1}{w}$ 1m

(iii) Find the Argument of $\frac{1}{w}$ 1m

b) If w is a complex root of the equation $z^3=1$, **note that** w^2 is the other complex root. Note also that. $1+w+w^2=0$ (No need to prove this)

Form a quadratic equation whose roots are $3+w$ and $3+w^2$ 3m

Question: 2

a) Given that $1+2i$ is a root of the polynomial $P(z)=z^4-2z^3+6z^2-2z+5$, factorise $P(z)$ in the Complex Number system. 4m

b) Given that $Q(x)=x^4+x^3-9x^2+11x-4$, has a root of multiplicity 3, factorise $Q(x)$. 4m

c) Express $\frac{x^3+2}{x^2-4}$ as a sum of partial fractions. 3m

Question 3

Sketch the locus of z : in each of the following: Clearly state the feature of each locus.

(i) $|z-i| = |z-2|$. 2m

(ii) $\frac{z-4i}{z-1}$ is purely imaginary Let $z=x+iy$ 3m

Question 4

α, β, γ are the roots of the polynomial equation $P(x)=0$,
where $P(x)=x^3-3x+1$,

write down the polynomial equation, whose roots are $\alpha^2+2, \beta^2+2, \gamma^2+2$ 3m

Question 5

(i) Solve for z : $z^7=1$, in the Complex Number System, where z is a complex number. 2m

(ii) Show that if w is a complex root of $z^7=1$,
 w^2, w^3, w^4, w^5, w^6 are the other complex roots.
Also identify the complex conjugates in these complex roots 2m

(iii) Factorise fully z^7-1 in the field of Real Numbers. 2m

(iv) Explain why $w, w^2, w^3, w^4, w^5, w^6$ are the roots of the equation
 $z^6+z^5+z^4+z^3+z^2+z+1=0$ 1m

(v) Show that $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=-\frac{1}{2}$ 2m

END OF PAPER

Q.1

$$z = 1+2i; w = 1+i$$

$$\begin{aligned} \textcircled{i} \quad z\bar{w} &= (1+2i)(1-i) \\ &= 1-2i^2-i+2i \\ &= 3+i \end{aligned}$$

$$\begin{aligned} \textcircled{ii} \quad \frac{1}{w} &= \frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1-i}{1-i^2} \\ &= \frac{1}{2} - \frac{1}{2}i \end{aligned}$$

$$\begin{aligned} \textcircled{iii} \quad \text{Arg of } \frac{1}{w} &= \text{Arg } \frac{1}{2} - \frac{1}{2}i \\ &= -\pi/4 \end{aligned}$$



$$\begin{aligned} \text{b) Sum of roots} & \text{ is } 3+w + 3+w^2 \\ &= 6+w+w^2 \\ &= 6-1=5 \quad (1+w+w^2=0, w+w^2=-1) \end{aligned}$$

$$\begin{aligned} \text{Product of roots} & \text{ is } (3+w)(3+w^2) \\ &= 9+3w+3w^2+w^3 \\ &= 10+3(w+w^2) \\ &= 10-3 \\ &= 7 \end{aligned}$$

$$\therefore \text{The equation is } x^2 - 5x + 7 = 0$$

Q.2

$$\begin{aligned} 1+2i & \text{ is a root; Coeffs of } P(z) \text{ are real} \\ 1-2i & \text{ is a root.} \end{aligned} \quad \textcircled{14}$$

$$\therefore (z-(1+2i))(z-(1-2i)) \text{ is a factor.}$$

$$z^2 - (2)z + (1-4i^2) \quad " \quad "$$

$$z^2 - 2z + 5 \text{ is a factor.} \quad \textcircled{14}$$

$$\begin{array}{r} z^2 - 2z + 5 \overline{) z^4 - 2z^3 + 6z^2 - 2z + 5} \\ \underline{z^4 - 2z^3 + 5z^2} \\ 1z^2 - 2z + 5 \\ \underline{1z^2 - 2z + 5} \\ 0 \end{array} \quad \textcircled{14}$$

$$\therefore P(z) = (z-(1+2i))(z-(1-2i))(z-i)(z+i) \quad \textcircled{14}$$

$$\begin{aligned} \textcircled{b} \quad Q(z) &= z^4 + z^3 - 9z^2 + 11z - 4 \text{ has a } \textcircled{1} \\ \text{if } z & \text{ is a root of multiplicity 3 for } Q(z) \\ " & " " " \quad " \quad 2 \text{ for } Q'(z) \\ " & " " " \quad " \quad 1 \text{ for } Q''(z) \end{aligned}$$

$$Q'(z) = 4z^3 + 3z^2 - 18z + 11 \quad \textcircled{2}$$

$$\begin{aligned} Q''(z) &= 12z^2 + 6z - 18 \\ &= 6(2z^2 + z - 3) \\ &= 6(2z+3)(z-1) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{It could be } 1 \text{ or } -\frac{3}{2} \quad Q'(1) &= 4+3-18+11 \\ &= 0 \\ Q(1) &= 1+1-9+11-4 \\ &= 0 \end{aligned}$$

$$\therefore a = 1$$

Q

$$Q(z) = (z-1)^3 (z+4) \quad \text{observation}$$

$$c) \quad \begin{array}{r} x \\ x^2-4 \overline{) x^3+2} \\ \underline{x^3-4x} \\ 4x+2 \end{array}$$

$$\therefore \frac{x^3+2}{x^2-4} = x + \frac{4x+2}{x^2-4}$$

$$= x + \frac{A}{x-2} + \frac{B}{x+2}$$

$$\therefore 4x+2 = A(x+2) + B(x-2)$$

$$x=2, \quad 10 = 4A; \quad A = \frac{5}{2}$$

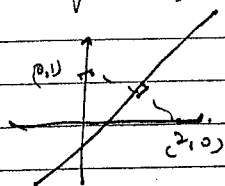
$$x=-2, \quad -6 = -4B \quad B = \frac{3}{2}$$

$$\therefore \frac{x^3+2}{x^2-4} = x + \frac{5}{2(x-2)} + \frac{3}{2(x+2)}$$

Q.3

$$|z-i| = |z-2|$$

$$\text{dist. from } (0,1) = \text{dist. from } (2,0)$$



Perpendicular bisector of the interval joining $(0,1)$ and $(2,0)$

11

$$\frac{z-4i}{z-1} = \frac{x+iy-4i}{(x-1)+iy} \cdot \frac{(x-1)-iy}{(x-1)-iy}$$

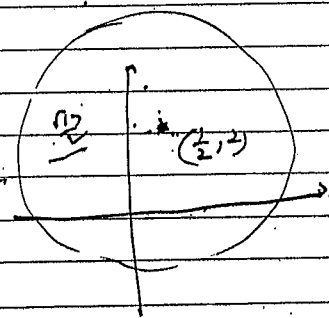
$$= \frac{x(x-1) + y(y-4) + i((x-1)(y-4) - xy)}{(x-1)^2 + y^2}$$

Purely Imaginary $\therefore$ Real part = 0

$$\begin{aligned} x(x-1) + y(y-4) &= 0 \\ x^2 - x + y^2 - 4y &= 0 \end{aligned}$$

$$(x-\frac{1}{2})^2 + (y-2)^2 = 4 + \frac{1}{4}$$

$$\text{Circle : Center } (\frac{1}{2}, 2) \\ r = \frac{\sqrt{17}}{2}$$



Q.4

$$P(x) = x^3 - 3x + 1$$

$$y = x^2 + 2, \quad x = \sqrt{y-2}$$

The required polynomial is

$$(\sqrt{y-2})^3 - 3\sqrt{y-2} + 1 = 0$$

$$\sqrt{y-2} (y-2-3) = -1$$

$$(y-2)(y-5)^2 = 1$$

$$\text{or } (x-2)(x-5)^2 = 1$$

(or find $Q(x)$ with roots $\alpha^2, \beta^2, \gamma^2$
 & then find the polynomial with roots $\alpha^2+2, \beta^2+2, \gamma^2+2$)

Q.5

$$z^7 = 1$$

$$\text{Let } z = r e^{i\theta}$$

$$|z| = r$$

$$|z^7| = 1$$

$$|z|^7 = 1$$

$$r^7 = 1$$

$$r = 1 \text{ (r is real)}$$

$$\therefore z = e^{i\theta}$$

$$z^7 = 1$$

$$e^{i7\theta} = 1 \quad (\text{De Moivre's Thm})$$

$$\cos 7\theta = 1; \sin 7\theta = 0$$

$$7\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, 12\pi$$

$$\theta = 0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}, \frac{8\pi}{7}, \frac{10\pi}{7}, \frac{12\pi}{7}$$

$\therefore$ The roots are

$$e^{i0}, e^{i\frac{2\pi}{7}}, e^{i\frac{4\pi}{7}}, e^{i\frac{6\pi}{7}}, e^{i\frac{8\pi}{7}}, e^{i\frac{10\pi}{7}}, e^{i\frac{12\pi}{7}}$$

$$\text{Let } \omega = e^{i\frac{2\pi}{7}}$$

$$e^{i\frac{4\pi}{7}} = \left(e^{i\frac{2\pi}{7}}\right)^2 \quad (\text{De Moivre's Thm})$$

$$= \omega^2$$

$$e^{i\frac{6\pi}{7}} = \left(e^{i\frac{2\pi}{7}}\right)^3 \quad (\text{ " })$$

$$= \omega^3$$

$$e^{i\frac{8\pi}{7}} = \left(e^{i\frac{2\pi}{7}}\right)^4 \quad (\text{ " })$$

$$= \omega^4$$

$$e^{i\frac{10\pi}{7}} = \omega^5 = \frac{\omega^5}{\omega^2}$$

$$e^{i\frac{12\pi}{7}} = \omega^6 = \frac{\omega^6}{\omega^2}$$

(iii)

$$z^7 - 1 = (z - 1)(z - \omega)(z - \bar{\omega})(z - \omega^2)(z - \bar{\omega}^2)$$

$$\text{more: } (z - \omega)(z - \bar{\omega}) = z^2 - (z\omega + z\bar{\omega}) + \omega\bar{\omega}$$

$$= (z - 1)(z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)$$

(2)

$$(z^2 - 2z \cos \frac{6\pi}{7} + 1)$$

(iv)

$$(z^7 - 1) = (z - 1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$$

(1)

These roots are $1, \omega, \omega^2, \omega^3, \omega^4, \omega^5, \omega^6$

$\omega, \omega^2, \omega^3, \omega^4, \omega^5, \omega^6$ are the roots of
 $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$

(v)

$$\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = -1$$

(2)

$$\text{or } \omega + \omega^2 + \omega^3 + \omega^3 + \omega^2 + \omega = -1$$

$$(\omega + \bar{\omega}) + (\omega^2 + \bar{\omega}^2) + (\omega^3 + \bar{\omega}^3) = -1$$

$$2\cos\frac{2\pi}{7} + 2\cos\frac{4\pi}{7} + 2\cos\frac{6\pi}{7} = -1$$

$$\therefore \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = \frac{-1}{2}$$